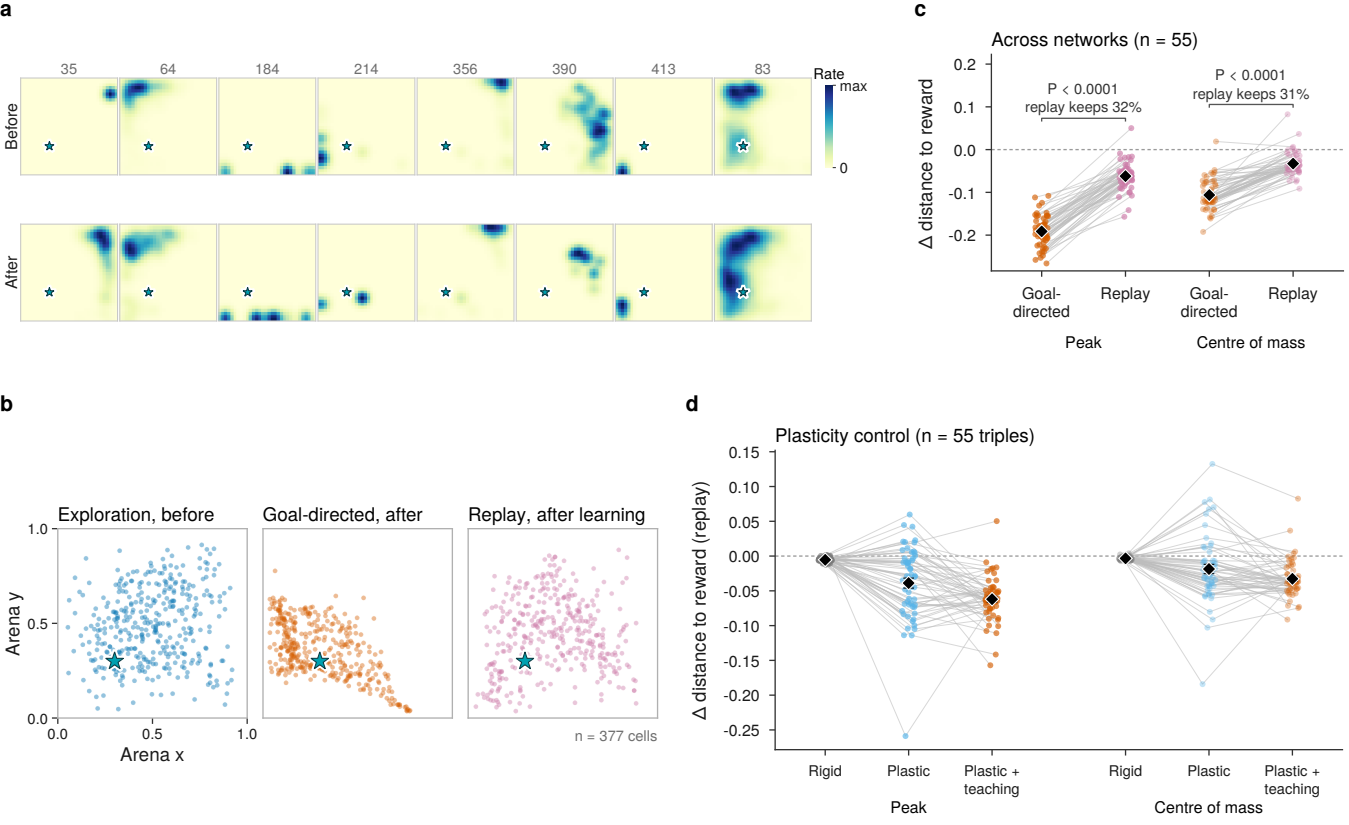

a
35
64
184
214
356
390
413
83
Rate
max
0
Before
After
c
Across networks (n = 55)
P < 0.0001
replay keeps 32%
P < 0.0001
replay keeps 31%
Δ distance to reward
0.2
0.1
0.0
-0.1
-0.2
Goal-directed
Replay
Goal-directed
Replay
Peak
Centre of mass
b
Exploration, before
Goal-directed, after
Replay, after learning
Arena y
Arena x
1.0
0.5
0.0
0.0
0.5
1.0
n = 377 cells
d
Plasticity control (n = 55 triples)
Δ distance to reward (replay)
0.15
0.10
0.05
0.00
-0.05
-0.10
-0.15
-0.20
-0.25
Rigid
Plastic
Plastic + teaching
Rigid
Plastic
Plastic + teaching
Peak
Centre of mass